

Whanos Documentation

Basic Description

Whanos's goal is to offer developper the opportunity to automatically containerize and deploy their application through a simple instance without having to bother about all the remaining steps.

A simple push and it's all magically up !!

Docker

Whanos support 5 langagues

- *C*
- *Java*
- *Javascript*
- *Befunge*
- *python*

A valid repository for the Whanos infrastructure is a project made with one of the languages above and following some basics rules: [refer to whanos project pdf](#)

There are two different types of images:

The standalone Images

```
FROM node:20.9
```

```
SHELL ["/bin/bash", "-c"]
```

```
COPY . /app
```

```
WORKDIR /app
```

```
RUN npm install
```

```
CMD ["node", "."]
```

The Base Images

```
FROM node:20.9
```

```
SHELL ["/bin/bash", "-c"]
```

```
ONBUILD COPY . /app
```

```
ONBUILD WORKDIR /app
```

```
ONBUILD RUN npm install
```

```
CMD ["node", "."]
```

The specificity being for app that runs without having to further configure their environment they'll just stick to the default standalone version while the other are going to reference the base image using a `FROM` statement.

Jenkins

Jenkins in the Whanos infrastructure is used to define sets of basic jobs like

- *Base Images build*
- *Project adding*
- *All base images build*

JCasC

It is configured using the plugin ***Configuration As Code***. Before deploying the instance you have to fill the `whanos/files/config/config.yaml` gap at the user admin password before running it so that it automatically creates a basic admin user.

```
jenkins:
```

```
  systemMessage: "Welcome Whanos folks\n\n"
```

securityRealm:

local:

allowsSignup: **false**

users:

- id: admin

name: Admin

password: // To be filled with chosen password before deploying the infrastructure

tool:

git:

installations:

- name: "git"

home: "/usr/bin/git"

security:

globalJobDslSecurityConfiguration:

useScriptSecurity: **false**

```
jobs:

  - file: "${JENKINS_HOME}/config/job_dsl.groovy"
```

It defines the different jobs using the **Job DSL** plugin in a groovy format.

Link project job

This job allow devs to link their project hosted on a git repository to the whanos infrastrucute. Every push made to the main branch will then trigger the job of the project which will directly push the image of the project to a docker registry and then deploy it using kubernetes

Parameters

- **GIT_URL**: Git URL starting from domain name upto repository name
- **GIT_USERNAME**: The username of the repository lead
- **GIT_PERSONAL_ACCESS_TOKEN**: A token giving access to every repository owned by *GIT_USERNAME*
- **DISPLAY_NAME**: Project name (will be used for image too)

Ansible

The whole infrastructure is deployed to a VM using ansible

```
[whanos]
WhanosMainVM      ansible_host=remote_host      ansible_connection=ssh      ansible_user=ssh_user
```

You have to add in there the `IP` of your VM as well as the name of your ssh user.

A simple

```
ansible-playbook -i production playbook.yaml
```

And it's setup and ready to be used

How to use

Navigate to the url of your jenkins instance

Go into the `link-projects` job and fill in the required information.

A job will be created in the `Projects` folder and it will be checking every minute for updates on your main branch.

When a change will be applied to the repository the job will be triggered and it will automatically be set into a docker image before being pushed to the docker registry hosted at `localhost:5000` of your VM

You're then free to use this image whenever you want into kubernetes to deploy it into a cluster.